

Gradient Norm Stability during MM-TTA

Note: Sine Adapter gradient inherently includes an $\omega/g \approx 10$ scaling factor via chain rule, yet strictly avoids explosion.

Gradient L2 Norm

10^2
 10^1
 10^0
 10^{-1}
 10^{-2}
 10^{-3}

0

10

20

30

40

50

Adaptation Steps

- Stable Gradient Regime
- Linear Adapter
- Sine Adapter (Ours)

